# Finite-$p$ Extrinsic Metrics as Moment-Constrained Coupling Problems

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**Abstract**

Bartošová, Lopez-Abad, Lupini, and Mbombo close their paper by saying that they do not know an explicit description of their three extrinsic metrics for finite $p$, analogous to their $p = \infty$ appendix. We give exact descriptions for every $1 \le p < \infty$. A finite family of maps from finite-dimensional spaces into $L_p[0, 1]$ is encoded exactly by the joint law of its coefficient-vector functions. Consequently the full-rank metric is a moment-constrained optimal coupling problem, the Kadets metric is a moment-constrained Hausdorff problem, and the square rank-$k$ metric is the infimum of the norm of an explicit finite-rank integral-kernel operator. The formulas contain only probability laws on finite-dimensional vector spaces and scalar $p$-moments. They also apply to the source's finite-support metrics through its Theorem 3.11.

## 1 Source question and precise answer

The source is D. Bartošová, J. Lopez-Abad, M. Lupini, and B. Mbombo, *The Ramsey properties for Grassmannians over* $\mathbb{R}, \mathbb{C}$, arXiv:1910.00311v1 [1]. It defines three extrinsic metrics: $\partial_{X,E}$ for norms represented by full-rank embeddings, $\gamma_{k,E}$ for isometry types of $k$-dimensional subspaces, and $d_{k,E}$ for $\mathrm{GL}(\mathbb{F}^k)$-orbits attached to rank-$k$ operators $E^* \to E$. After giving concrete descriptions when $p = \infty$, the authors write on PDF page 24:

> "We do not know a similar explicit description of the extrinsic metrics for $p$'s other than $\infty$."

See Figure 1.

It follows from the inequality in (6) and Proposition A.1 *b)* that

$$\mathfrak{d}_{k,\ell_\infty^\infty}([\mathbf{m}],[\mathbf{n}]) \leqslant \|T-U\|_{\ell_1^\infty,\ell_\infty^\infty} \leqslant k^2\lambda^3(\omega(\mathtt{m}_1,\Delta\cdot\mathtt{n}_1)+\omega(\mathtt{m}_0,\Delta\cdot\mathtt{n}_0)) \leqslant k^2\lambda^3\widetilde{\omega}_2([\mathbf{m}],[\mathbf{n}]). \quad \square \quad \square$$

We do not know a similar explicit description of the extrinsic metrics for $p$'s other than $\infty$.

Figure 1: The closing question in arXiv:1910.00311v1, PDF p. 24.

We interpret "description" as an exact variational formula that eliminates the ambient infinite-dimensional embeddings. Under that precise and natural interpretation, Theorem 2 gives a full answer for all finite $p$. It is a probabilistic finite-dimensional formula, rather than a claim that the metrics always reduce to elementary closed forms.

## 2 Proof intuition

Choose a basis of a finite-dimensional space $W$. Every bounded map $A : W \to L_p[0,1]$ has the pointwise form

$$(Ax)(\omega) = a(\omega)(x), \qquad a(\omega) \in W^*.$$

For several maps into the same $L_p$ space, retain the joint law of their coefficient vectors. The condition that $A$ is an isometric embedding is then exactly a list of moment identities

$$\int |a(x)|^p \, d\mu(a) = \|x\|^p \quad (x \in W).$$

Conversely, every such law can be realized by a random coefficient vector on $[0,1]$. Thus no information relevant to simultaneous embeddings is lost when the maps are replaced by a probability coupling.

For $\partial$, the norm of the difference of two embeddings becomes the supremum of one scalar $p$-moment. For $\gamma$, the two subspace unit balls become two coordinate unit balls inside the common seminorm induced by the coupling, so their opening is an ordinary Hausdorff formula. For $d_k$, write $T = T_0 T_1^*$. If $a_0(\omega) \in V^*$ and $a_1(\eta) \in V$ are the two coefficient vectors, then $T$ is represented by the kernel $a_0(\omega)(a_1(\eta))$. Subtracting two factorizations gives the kernel in Theorem 2(c).

## 3 Coefficient laws

Throughout, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, $1 \le p < \infty$, and $q \in (1, \infty]$ is conjugate to $p$. All probability laws below are Borel laws with the displayed finite $p$-moments.

**Lemma 1** (Joint coefficient-law lemma)**.** *Let $W_1, \dots, W_s$ be finite-dimensional $\mathbb{F}$-spaces. Simultaneous bounded maps $A_j : W_j \to L_p[0,1]$ determine a Borel probability law $\mu$ on $W_1^* \times \cdots \times W_s^*$ such that, if $z_j$ is the $j$th coordinate,*

$$\|A_j x\|_p^p = \int |z_j(x)|^p \, d\mu(z) \qquad (x \in W_j). \tag{1}$$

*Conversely, every such law is realized by simultaneous maps on $L_p[0,1]$, with* (1)*. Moreover, all joint pointwise linear combinations of the maps have exactly the corresponding moments under $\mu$.*

*Proof.* For a basis $(e_i)$ of $W_j$, choose measurable representatives of $A_j e_i$ and put

$$a_j(\omega) = \sum_i (A_j e_i)(\omega) e_i^*.$$

Then $(A_j x)(\omega) = a_j(\omega)(x)$ outside one common null set. The joint push-forward of Lebesgue measure by $(a_1, \dots, a_s)$ is $\mu$. Finite dimensionality and the fact that every coordinate belongs to $L_p$ give the required finite moments, and integration proves (1) and the final assertion.

Conversely, a Borel probability law on a finite-dimensional real or complex vector space is the law of a Borel random variable on $[0,1]$ (the standard randomization theorem; see, for example, [2]). Realize the coordinate tuple $z(\omega)$ and define $(A_j x)(\omega) = z_j(\omega)(x)$. The moment assumption puts these functions in $L_p$, and (1) follows directly. □

Let $V$ be $k$-dimensional. For norms $m, n$ on $V$, let $\mathcal{C}_p(m,n)$ be the laws $\mu$ of pairs $(a,b) \in V^* \times V^*$ satisfying

$$\int |a(x)|^p \, d\mu = m(x)^p, \qquad \int |b(x)|^p \, d\mu = n(x)^p \quad (x \in V). \tag{2}$$

Set
$$\rho_\mu(x,y) = \left( \int |a(x) - b(y)|^p \, d\mu(a,b) \right)^{1/p}.$$
For a reference norm $r$ on $V$, define
$$D_r(\mu) = \sup_{r(x)\le 1} \rho_\mu(x,x),$$
and define
$$G_{m,n}(\mu) = \max \left\{ \sup_{m(x)\le 1} \inf_{n(y)\le 1} \rho_\mu(x,y), \right. \\ \left. \sup_{n(y)\le 1} \inf_{m(x)\le 1} \rho_\mu(x,y) \right\}. \tag{3}$$

For the square metric, a law $\mu$ is now on
$$\mathcal{Z}_V = V^* \times V \times V^* \times V, \qquad z = (a_0, a_1, b_0, b_1).$$
It induces norms
$$m_{0,\mu}(x) = \left( \int |a_0(x)|^p \, d\mu \right)^{1/p}, \qquad m_{1,\mu}(f) = \left( \int |f(a_1)|^p \, d\mu \right)^{1/p},$$
$$n_{0,\mu}(x) = \left( \int |b_0(x)|^p \, d\mu \right)^{1/p}, \qquad n_{1,\mu}(f) = \left( \int |f(b_1)|^p \, d\mu \right)^{1/p}.$$
For $\mathbf{m}, \mathbf{n} \in \mathcal{D}_k(L_p)$, let $\mathcal{Q}_p(\mathbf{m}, \mathbf{n})$ consist of those $\mu$ for which these are genuine norms and
$$[(m_{0,\mu}, m_{1,\mu})] = \mathbf{m}, \qquad [(n_{0,\mu}, n_{1,\mu})] = \mathbf{n}$$
in the source's diagonal $\mathrm{GL}(V)$-orbit space. On two independent copies $z = (a_0, a_1, b_0, b_1)$ and $z' = (a_0', a_1', b_0', b_1')$, put
$$K_\mu(z, z') = a_0(a_1') - b_0(b_1') \tag{4}$$
and
$$(\mathsf{K}_\mu g)(z) = \int K_\mu(z,z') g(z') \, d\mu(z'). \tag{5}$$
We use the complex bilinear realization of the duality $L_p^* = L_q$, namely $\langle f, g \rangle = \int fg$. With the usual sesquilinear convention, one conjugates $g$ in (5); the operator norm is the same.

**Theorem 2** (Exact finite-$p$ descriptions)**.** *For every $1 \le p < \infty$ the following hold.*

(a) *If $X = (V, r)$ and $m, n \in \mathcal{N}_V(L_p)$, then*
$$\boxed{\partial_{X,L_p}(m,n) = \inf_{\mu \in \mathcal{C}_p(m,n)} D_r(\mu).} \tag{6}$$

(b) *If $[m], [n] \in \mathcal{B}_k(L_p)$, then*
$$\boxed{\gamma_{k,L_p}([m],[n]) = \inf_{\mu \in \mathcal{C}_p(m,n)} G_{m,n}(\mu).} \tag{7}$$

(c) *If* $\mathbf{m}, \mathbf{n} \in \mathcal{D}_k(L_p)$*, then*

$$\boxed{d_{k,L_p}(\mathbf{m}, \mathbf{n}) = \inf_{\mu \in \mathcal{Q}_p(\mathbf{m},\mathbf{n})} \|\mathsf{K}_\mu\|_{L_q(\mu) \to L_p(\mu)}.} \tag{8}$$

*The formulas are invariant under changing representatives in the indicated orbit classes.*

*Proof.* For (a), take simultaneous isometric embeddings $T : (V, m) \to L_p$ and $U : (V, n) \to L_p$. Lemma 1 gives a law $\mu \in \mathcal{C}_p(m, n)$, and pointwise

$$((T - U)x)(\omega) = a(\omega)(x) - b(\omega)(x).$$

Therefore $\|T - U\|_{(V,r) \to L_p} = D_r(\mu)$. Conversely every admissible $\mu$ produces exactly such simultaneous isometric embeddings. Taking the infimum on the two equivalent parameter spaces proves (6).

For (b), the unit balls in the ranges $T(V)$ and $U(V)$ are respectively

$$\{Tx : m(x) \le 1\}, \qquad \{Uy : n(y) \le 1\}.$$

Their ambient $L_p$ distance satisfies $\|Tx - Uy\|_p = \rho_\mu(x, y)$. Thus their Hausdorff distance is exactly (3). Lemma 1 again identifies all simultaneous realizations with all laws in $\mathcal{C}_p(m, n)$, giving (7). Replacing $m$ or $n$ by an isometric representative merely reparametrizes $x$ or $y$, which also proves the asserted invariance here.

For (c), first take admissible factorizations

$$T = T_0 T_1^*, \qquad U = U_0 U_1^*,$$

where $T_0, U_0 : V \to L_p$ and $T_1, U_1 : V^* \to L_p$ are injective and induce the prescribed orbit pairs. Apply Lemma 1 to the four maps. Their joint law lies in $\mathcal{Q}_p(\mathbf{m}, \mathbf{n})$. In the coefficient notation above, for $g \in L_q(\mu)$,

$$T_1^* g = \int a_1(z') g(z') \, d\mu(z'), \qquad (T_0 T_1^* g)(z) = \int a_0(a_1') g(z') \, d\mu(z').$$

The analogous identity for $U$ shows that $T - U = \mathsf{K}_\mu$, including equality of the $L_q \to L_p$ operator norms.

Conversely, any $\mu \in \mathcal{Q}_p(\mathbf{m}, \mathbf{n})$ can be realized on $[0, 1]$. Its four coordinates define $T_0, T_1, U_0, U_1$ by evaluation. The moment norms being genuine norms makes all four maps injective. In finite dimension, the adjoints $T_1^*, U_1^*$ are onto $V$, so both products have rank $k$, and the orbit constraints make the factorizations admissible for the source definition. Formula (5) again gives $T - U = \mathsf{K}_\mu$. Taking infima proves (8). Finally, the use of orbit constraints rather than fixed representatives makes the $\mathrm{GL}(V)$ invariance in (c) automatic. $\square$

**Corollary 3** (The source metrics)**.** *Let* $E_p = (\mathbb{F}^\infty, \|\cdot\|_p)$ *and let* $\widehat{E}_p = L_p[0, 1]$ *be its Fraïssé limit in the notation of the source. On the domains* $\mathcal{N}_V(E_p)$*,* $\mathcal{B}_k(E_p)$*, and* $\mathcal{D}_k(E_p)$*, the three* $E_p$*-extrinsic metrics have the respective descriptions* (6)*,* (7)*, and* (8)*. The same formulas describe the metrics on the full limit-space domains.*

*Proof.* The source identifies the finite-$p$ limit space with $L_p[0, 1]$ and proves in its Theorem 3.11 that

$$\partial_{X,E_p} = \partial_{X,\widehat{E}_p}, \qquad \gamma_{k,E_p} = \gamma_{k,\widehat{E}_p}, \qquad d_{k,E_p} = d_{k,\widehat{E}_p}$$

on the corresponding finite-support domains [1, Theorem 3.11]. Apply Theorem 2 with $\widehat{E}_p = L_p[0, 1]$. $\square$

## 4 Verification and edge cases

The proof is exact and does not depend on numerical evidence. The script `code/finite_atomic_checks.py` generates 100 random finite atomic coefficient laws. It independently constructs the corresponding weighted $\ell_2$ embeddings and the square-operator matrices, then checks the moment identities and the equality between the factorized operator and the kernel operator. The command

```
conda run -no-capture-output -n sandbox python \
    code/finite_atomic_checks.py -trials 100
```

completed with maximum residual below $10^{-12}$. This is a regression check of the formulas, not a proof.

For $p = 1$, $q = \infty$ and the same finite-rank kernel calculation is valid; no reflexivity is used. For complex scalars, the bilinear and customary sesquilinear $L_p$ duality identifications differ only by conjugation. For $1 < p < \infty$ there are no additional restrictions. Although the source's Ramsey factorization theorems exclude certain even $p \geq 4$, the metric description itself uses only the coefficient-law lemma and therefore holds for every finite $p$.

## 5 Novelty, limitations, and review recommendation

The bounded novelty check on July 21, 2026 searched the run registry, solution and attempt indexes, the local parsed arXiv corpus, the exact source sentence, the paper title together with "extrinsic metrics" and $L_p$, the four authors' names with those terms, and "$E$-Kadets metric" with $L_p$. The arXiv/web searches returned the original arXiv:1910.00311 and unrelated or adjacent papers, but no later paper claiming these formulas or an answer to the closing question. Novelty confidence is therefore moderate, not definitive.

The main limitation is terminological: the source does not formally define what degree of simplification counts as an "explicit description." The present formulas are exact and eliminate optimization over maps into an infinite-dimensional ambient space, replacing it by moment-constrained laws on fixed finite-dimensional coefficient spaces. They are not generally closed elementary expressions in the input norms.

Status: `candidate full solution - likely valid`, subject to human review. The most important review points are (i) whether this precise variational meaning of "explicit" matches the authors' intended question, (ii) the complex duality convention in the kernel formula, and (iii) the application of the source's Theorem 3.11 to the three finite-support domains.

## References

[1] D. Bartošová, J. Lopez-Abad, M. Lupini, and B. Mbombo, *The Ramsey properties for Grassmannians over* $\mathbb{R}$, $\mathbb{C}$, arXiv:1910.00311v1 (2019).

[2] O. Kallenberg, *Foundations of Modern Probability*, 2nd ed., Springer, 2002.